

FUKUOKA AP, Japan

WMO: 478080

Lat: 33.584N

Lon: 130.452E

Elev: 40

StdP: 14.67

Time Zone: 9.00 (E09)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	31.9	33.8	16.0	12.4	39.1	19.0	14.3	40.3	22.7	39.9	20.9	41.5	7.7	150	0.436

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	12.2	93.3	78.3	91.5	77.9	89.6	77.2	80.1	88.6	79.3	87.5	78.6	86.4	11.2	310

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
77.4	142.6	84.4	77.0	140.9	84.1	75.6	134.2	83.4	43.8	89.3	42.8	88.1	42.0	86.3	83.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%		2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
Min		Max		Min	Max		Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)		(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
20.7	18.4	16.6	DB	28.0	96.5	1.9	1.5	26.7	97.6	25.6	98.5	24.5	99.3	23.2	100.4
			WB	25.1	81.3	2.1	1.2	23.6	82.2	22.4	82.9	21.2	83.6	19.6	84.5

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	62.9	43.2	45.0	50.7	59.4	67.9	74.4	81.9	83.6	76.7	67.1	56.8	47.2
	DBStd	14.69	4.62	5.53	5.66	5.51	4.33	4.03	4.15	3.42	4.59	4.88	5.71	5.24
	HDD50	563	217	159	58	3	0	0	0	0	0	0	8	118
	HDD65	2716	676	560	443	181	20	0	0	0	0	33	252	551
	CDD50	5283	6	19	81	285	554	732	990	1043	800	529	212	32
	CDD65	1961	0	0	0	13	110	282	525	578	350	97	6	0
	CDH74	17826	0	0	1	47	493	1639	5713	6824	2720	379	10	0
	CDH80	6413	0	0	0	3	62	364	2270	2935	743	36	0	0
Wind	WSAvg	7.8	8.2	8.0	8.5	8.3	7.8	7.2	8.1	7.9	7.0	6.7	7.1	8.1
Precipitation	PrecAvg	64.8	2.9	2.8	3.9	4.9	5.6	10.0	10.5	7.2	7.1	3.7	3.4	2.5
	PrecMax	117.3	8.2	6.1	8.8	9.4	19.5	25.3	34.9	33.3	24.0	11.4	7.2	5.8
	PrecMin	35.2	0.5	0.7	0.8	1.4	1.5	0.6	0.7	0.3	0.9	0.4	0.5	0.6
	PrecStd	14.5	1.5	1.3	1.7	2.0	3.1	5.2	7.2	5.7	4.3	2.5	1.7	1.2
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	61.2	66.6	71.7	78.7	84.5	89.7	94.9	96.7	91.8	83.8	75.3	64.7
		MCWB	54.5	58.1	56.9	64.4	65.8	76.0	78.7	78.6	77.3	72.0	63.9	56.2
	2%	DB	57.0	61.2	67.6	75.0	82.0	86.3	92.8	93.6	89.3	80.6	71.6	61.1
		MCWB	50.1	54.2	56.2	61.9	65.7	73.5	78.3	78.3	76.4	69.2	62.1	53.4
	5%	DB	53.8	57.4	64.0	71.8	79.0	84.1	91.2	91.8	86.3	78.4	68.4	59.0
		MCWB	47.1	50.3	54.8	59.5	64.3	72.3	77.9	78.1	74.8	67.7	59.9	51.7
	10%	DB	51.8	55.1	60.8	69.4	76.8	82.1	89.4	91.0	84.2	75.5	66.4	56.9
		MCWB	45.3	48.3	52.7	58.6	64.0	71.6	77.4	77.8	73.8	65.5	58.8	50.3
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	56.7	59.8	61.8	69.1	72.9	78.5	81.0	81.6	79.5	74.5	66.9	59.4
		MCDB	60.0	64.2	66.3	73.8	77.7	84.8	89.8	90.2	86.7	79.9	71.6	62.0
	2%	WB	52.3	55.8	58.7	64.8	70.6	76.8	80.0	80.4	78.1	72.1	64.0	55.7
		MCDB	55.7	60.6	64.3	71.1	75.8	82.9	88.5	89.1	85.1	78.2	68.7	59.4
	5%	WB	48.6	52.2	56.0	62.5	68.4	75.4	79.2	79.7	76.8	69.8	62.0	53.0
		MCDB	52.5	55.9	61.9	68.6	74.5	81.4	87.3	88.2	83.8	75.4	67.2	57.6
	10%	WB	45.8	49.1	53.9	60.2	66.3	73.7	78.4	78.9	75.6	67.4	60.0	50.7
		MCDB	50.6	53.8	60.1	66.4	73.2	79.5	86.2	87.2	82.4	73.3	65.5	55.8
Mean Daily Temperature Range	5% DB	MDBR	12.0	13.0	14.1	14.9	14.8	11.9	11.3	12.2	12.4	14.0	14.1	12.8
		MCDBR	16.2	17.6	19.0	19.1	18.6	15.1	13.5	14.4	14.4	16.4	16.3	16.1
		MCWBR	11.4	12.4	11.6	9.9	8.3	6.6	5.0	5.3	5.9	7.8	9.2	10.9
	5% WB	MCDBR	13.8	14.4	15.1	13.7	13.0	11.0	12.2	12.7	12.3	11.8	13.1	13.5
		MCWBR	11.9	12.3	11.4	9.5	8.7	6.2	5.3	5.3	5.7	7.0	9.2	11.0
Clear-Sky Solar Irradiance	taub	0.433	0.474	0.536	0.556	0.563	0.578	0.570	0.537	0.507	0.461	0.433	0.425	
	taud	2.047	1.973	1.827	1.833	1.875	1.923	1.989	2.077	2.112	2.141	2.132	2.082	
	Ebn at Noon	236	240	235	239	238	234	235	241	242	242	235	230	
	Edh at Noon	43	51	64	66	64	61	57	51	47	43	39	39	
All-Sky Solar Radiation	RadAvg	687	912	1259	1522	1654	1408	1535	1618	1300	1103	820	637	
	RadStd	65	90	85	129	190	174	228	177	133	115	78	43	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
Station Only		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (29 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page